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ЛАБОРАТОРНАЯ РАБОТА №2
“Приложения машинного обучения”

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1 Запущенный сервис

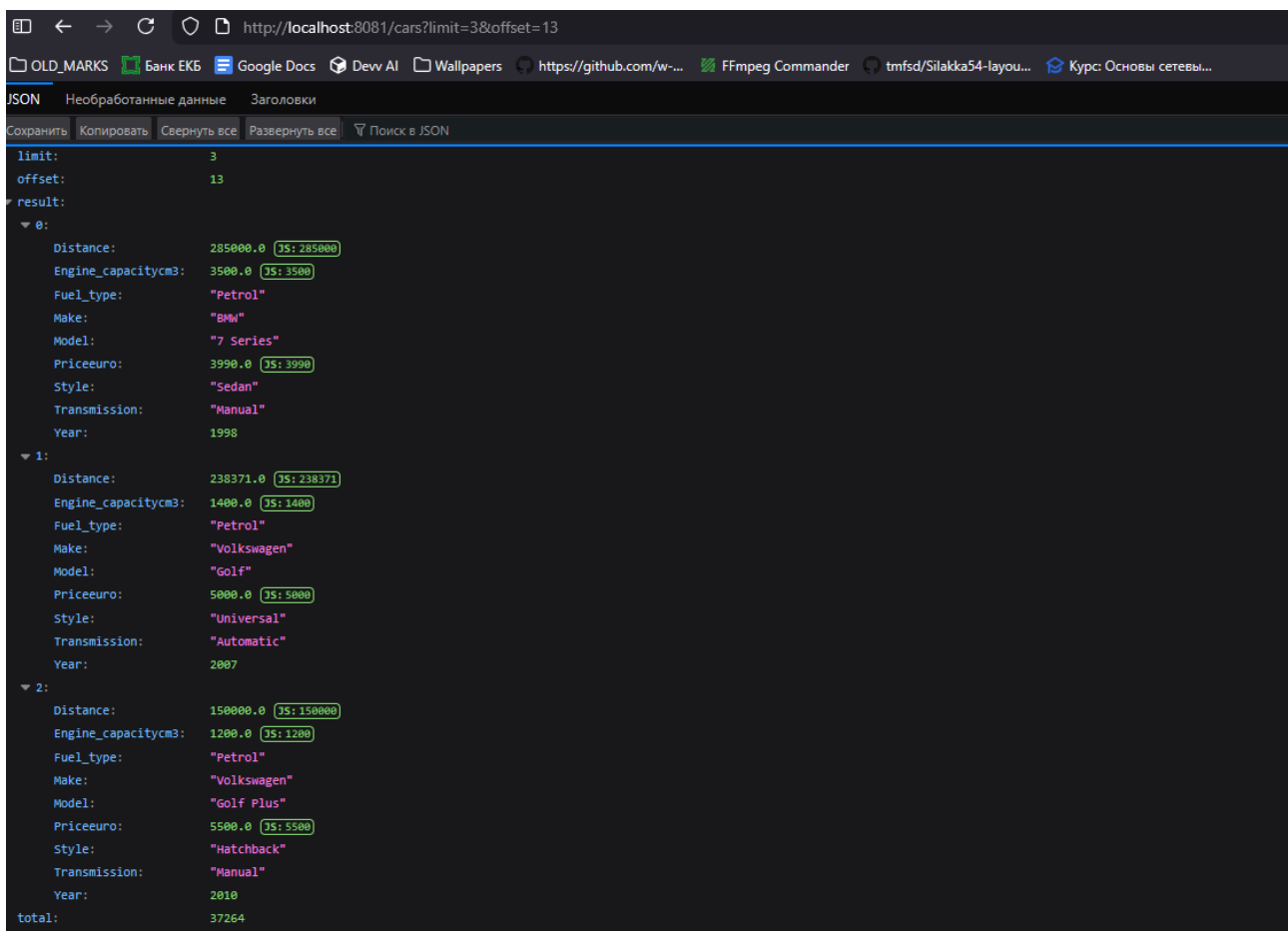
1.1 Сборка контейнеров

```
..ps/MLOPS/cars  x  +  v
cars:~$ compose.yaml  15s  10s  pyproject.toml  51ms  uv.lock
+ cars git:(main) X docker compose up -d --build
[+] Building 6.2s (13/13) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 528B 0.0s
=> [internal] load build definition from Dockerfile 0.2s
=> => transferring dockerfile: 380B 0.1s
=> [internal] load metadata for docker.io/library/python:3.11-slim 2.1s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.4s
=> => transferring context: 2B 0.3s
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:d6e4d224f70f9e0172a06a3a2eba2f768eb146811a349278b38fff3a 0.0s
=> [internal] load build context 0.3s
=> => transferring context: 93B 0.2s
=> CACHED [2/5] COPY requirements.txt /tmp/requirements.txt 0.0s
=> CACHED [3/5] RUN apt-get update && apt-get install unzip && rm -rf /var/lib/apt/lists/* 0.0s
=> CACHED [4/5] RUN pip install -r /tmp/requirements.txt && rm -f /tmp/requirements.txt 0.0s
=> CACHED [5/5] COPY app.py cars.csv / 0.0s
=> exporting to image 0.1s
=> => exporting layers 0.0s
=> => writing image sha256:e5cc4c3d7616a666b9fdefde7b9ac3056c4ee80613bf4810aaa4ad06c56d3198 0.0s
=> => naming to docker.io/library/cars-carsapi 0.0s
=> resolving provenance for metadata file 0.0s
[+] up 5/5
✓Image cars-carsapi Built 7.8sss
✓Container cars-redis-1 Healthy 146.6s
✓Container cars-postgres-1 Healthy 146.6s
✓Container cars-airflow-init-1 Exited 141.3s
✓Container cars-airflow-apiserer-1 Healthy 121.9s
+ cars git:(main) X
```

1.2 Работа контейнеров

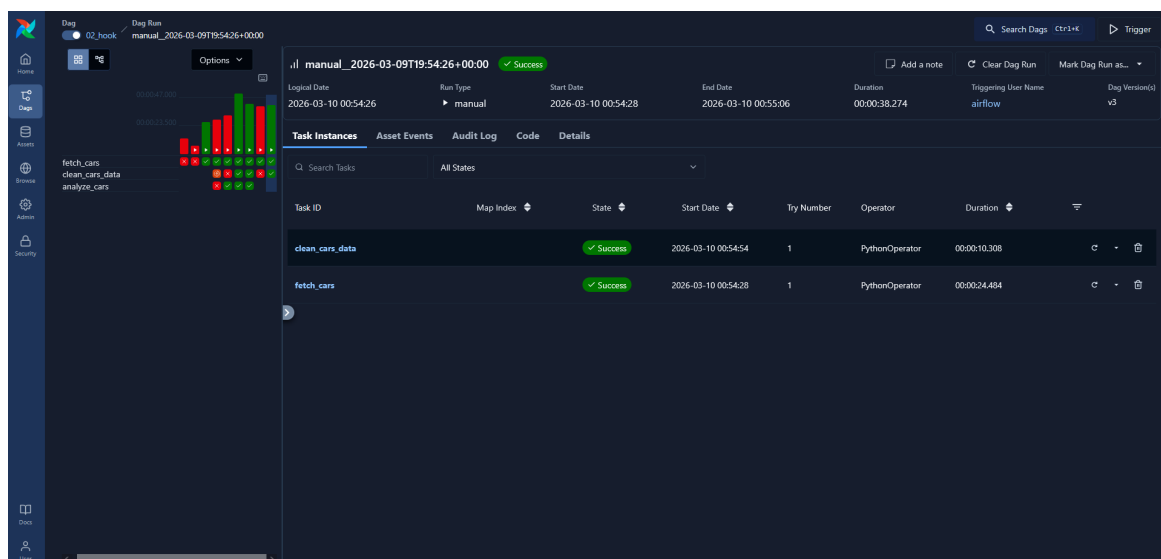
```
..ps/MLOPS/cars  x  +  v
=> => writing image sha256:e5cc4c3d7616a666b9fdefde7b9ac3056c4ee80613bf4810aaa4ad06c56d3198 0.0s
=> => naming to docker.io/library/cars-carsapi 0.0s
=> resolving provenance for metadata file 0.0s
[+] up 5/5
✓Image cars-carsapi Built 7.8sss
✓Container cars-redis-1 Healthy 146.6s
✓Container cars-postgres-1 Healthy 146.6s
✓Container cars-airflow-init-1 Exited 141.3s
✓Container cars-airflow-apiserer-1 Healthy 121.9s
+ cars git:(main) X docker compose ps
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
cars-airflow-apiserer-1 apache/airflow:3.1.2 "/usr/bin/dumb-init ..." airflow-apiserer 2 hours ago Up 2 hours (healthy) 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
cars-airflow-dag-processor-1 apache/airflow:3.1.2 "/usr/bin/dumb-init ..." airflow-dag-processor 2 hours ago Up 2 hours (healthy) 8080/tcp
cars-airflow-scheduler-1 apache/airflow:3.1.2 "/usr/bin/dumb-init ..." airflow-scheduler 2 hours ago Up 2 hours (healthy) 8080/tcp
cars-airflow-triggerer-1 apache/airflow:3.1.2 "/usr/bin/dumb-init ..." airflow-triggerer 2 hours ago Up 2 hours (healthy) 8080/tcp
cars-airflow-worker-1 apache/airflow:3.1.2 "/usr/bin/dumb-init ..." airflow-worker 2 hours ago Up About an hour (healthy) 8080/tcp
cars-carsapi-1 cars-carsapi "python /app.py" carsapi 2 hours ago Up 2 hours 0.0.0.0:8081->8081/tcp, [::]:8081->8081/tcp
cars-postgres-1 postgres:16 "docker-entrypoint.s..." postgres 2 hours ago Up 2 hours (healthy) 5432/tcp
cars-redis-1 redis:7.2-bookworm "docker-entrypoint.s..." redis 2 hours ago Up 2 hours (healthy) 6379/tcp
```

1.3 Работа API



2 Работа DAG

2.1 Работа пайплайна



2.2 Оператор `clean_cars_data`

В ходе лабораторной работы был добавлен оператор `clean_cars_data`. Код модифицированного пайплайна доступен в репозитории GitHub. Оператор выполняет все требуемые действия:

1. Очистка объектов с пропусками
2. Очистка дубликатов
3. Преобразование категориальных признаков в числовые